STUDENT DROPOUT PREDICTIVE MODEL

A Machine Learning Approach to Understanding and Predicting Student Attrition in Kenyan Universities

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Agenda

01. Project Background & Problem Statement
02. Data Overview & Processing
03. Exploratory Data Analysis (EDA)
04. Predictive Modelling Approach
05. Model Performance & Evaluation
06. Key Findings & Actionable Recommendations
07. Conclusion & Next Steps

Project Background

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Problem Statement

The Challenge:

Student dropout from higher education is a critical issue in Kenya, leading to:

i. Wasted educational resources for institutions.

ii. A reduction in the skilled workforce.

iii. Increased unemployment

iv. Lower lifetime earnings for individuals.

The Goal:

To understand the complex, multi-faceted factors (academic, financial, demographic) that contribute to this phenomenon and **build a predictive model to identify at-risk students early.**

NOTE: *We are not just predicting dropout, we are enabling early intervention.*

Problem Identification

Question 1.

What is the primary driver of dropout: academic preparedness, current performance, or financial constraints?

Question 2.

How do financial factors interact with academic factors to influence dropout risk?

Question 3.

Does the impact vary significantly by demographics, programme of study, or engagement levels?

Question 4.

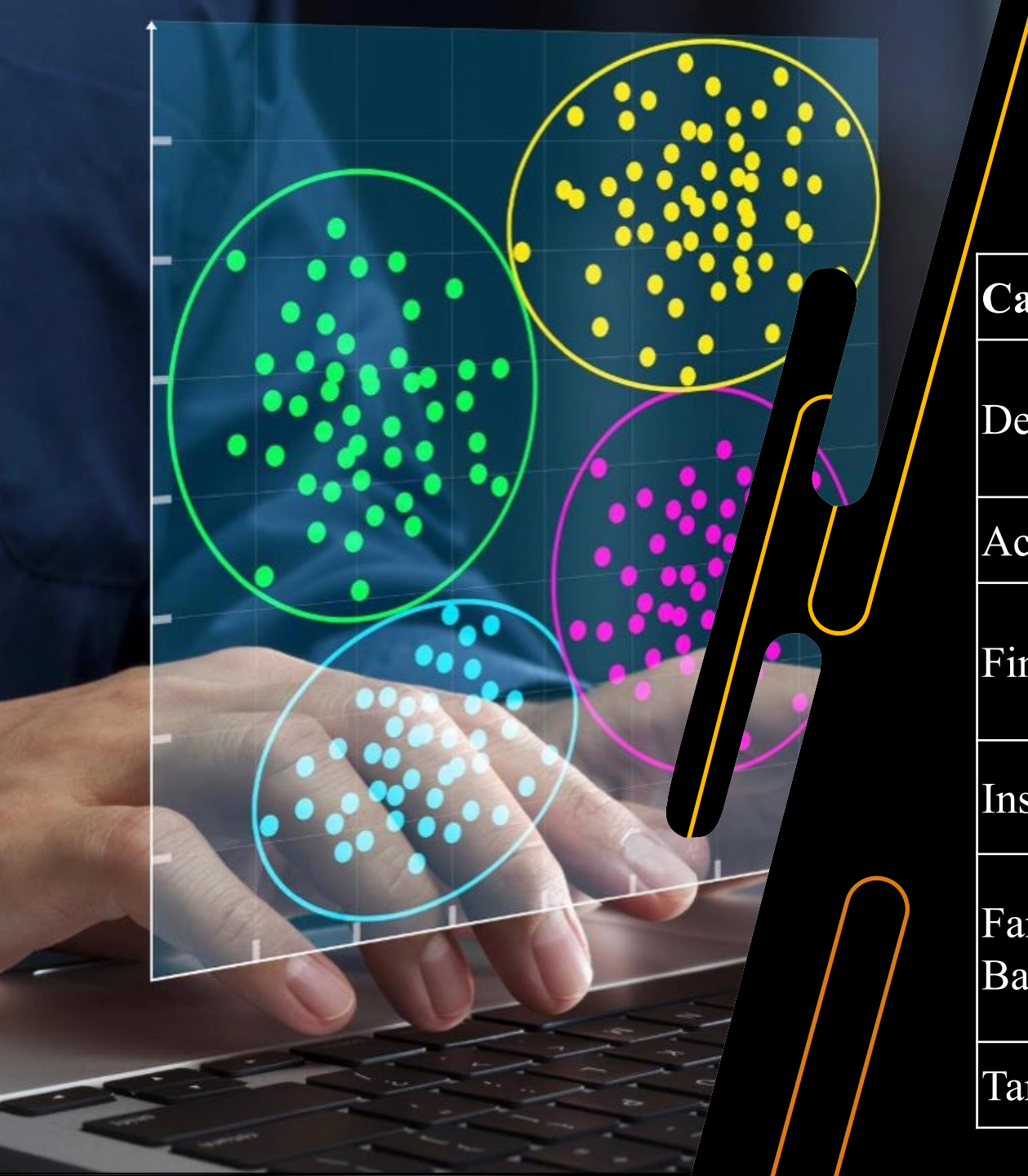
Can a predictive model accurately identify high-risk students?

Data Overview

Dataset: Student-level records from Kenyan higher education institutions.

Data Source: Higher Education Loans Board

Size: Over 118,000 students with 28 features.



Variable Categories

Category	Variables
Demographic	Gender, Age, County
Academic	Exam year, Course category
Financial	Program Cost, Total Loan Allocated, Scholarship Applied, Loan Status
Institutional	InstitutionName, UniversityType, Sponsored
Family Background	Mother_Age, Father_Age, Father_educ_level, Mother_educ_level, Highest_educ_level
Target	Drop out

Data Cleaning & Feature Engineering:

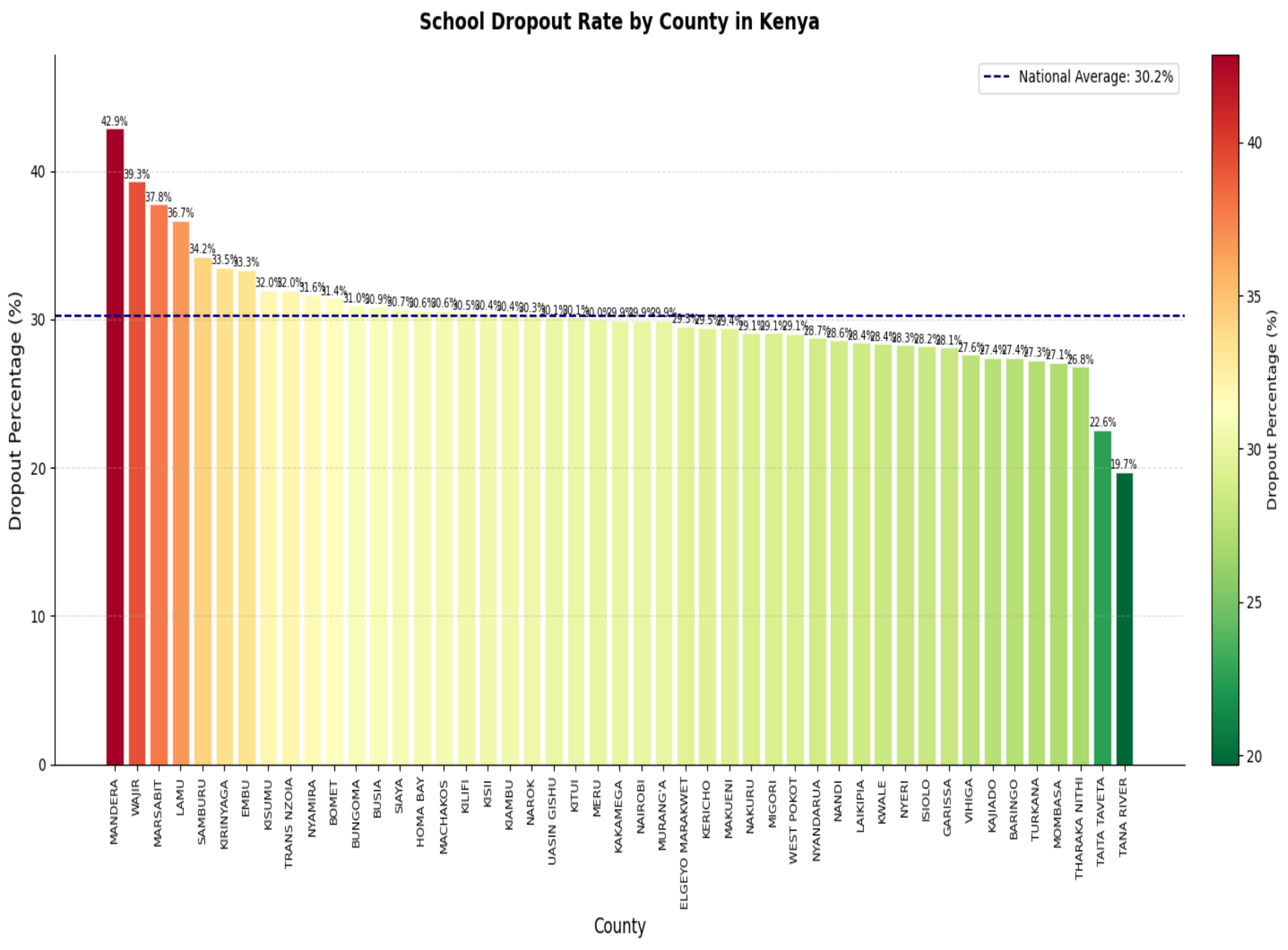
i. Dropped irrelevant/unusable columns (e.g., Bursary Batch Number, Index number).

ii. Removed rows with missing values in key columns.

iii. Created new features: Loan To Cost Ratio, Enrollment Age, Age Group.

iv. Focused final analysis on students enrolled in **2024**, resulting in **39,166** records with a **dropout rate of ~30.15%.**

Exploratory Data Analysis



Dropout by County

Key Insights:

✓ Highest Dropout Rates

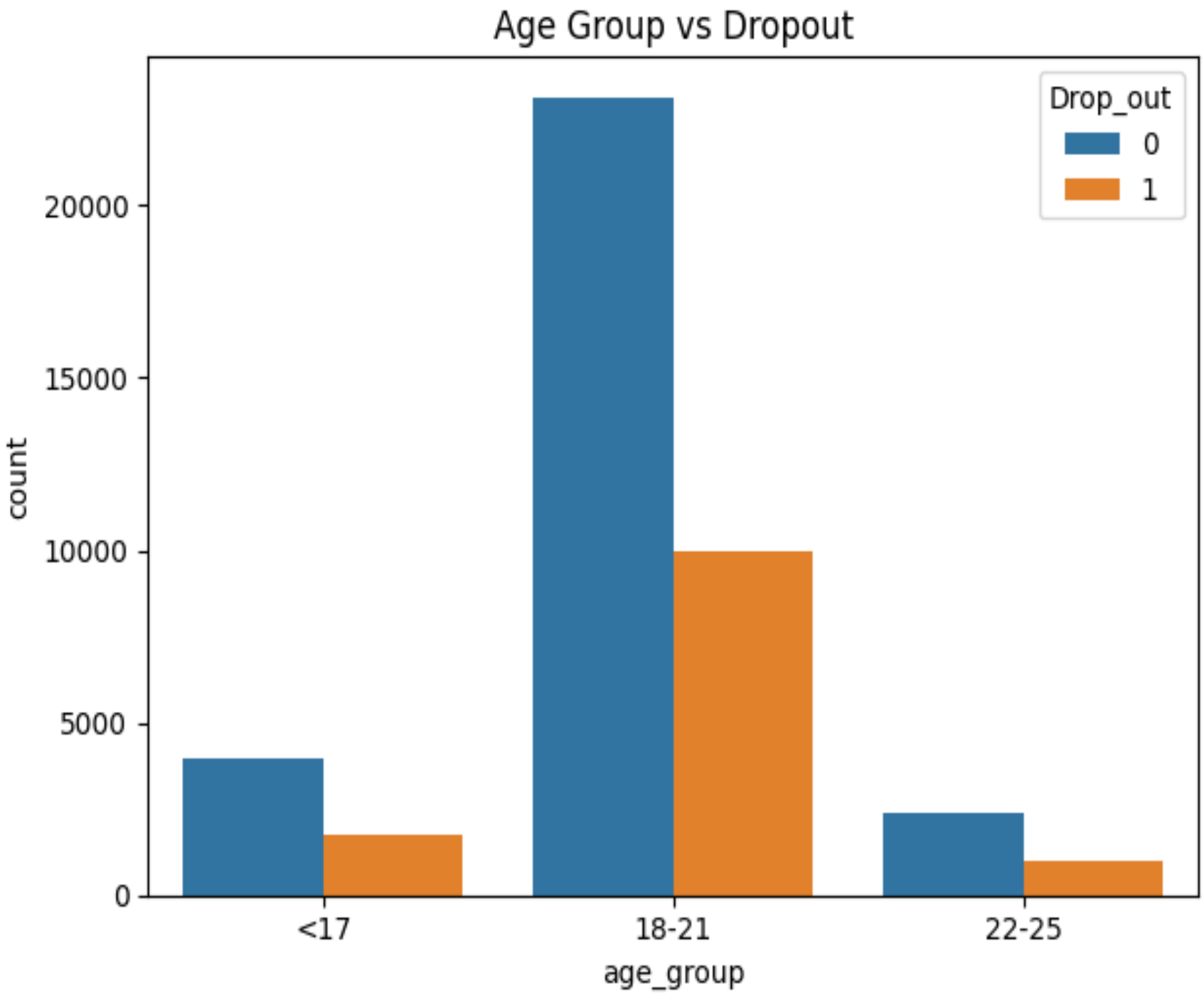
Counties like **Mandera (42.9%)**, **Wajir (39.3%)**, and **Marsabit (37.8%)** show the highest attrition.

✓ Regional Disparities:

A clear geographical pattern exists, with many counties in the northern and eastern regions showing above-average dropout rates.

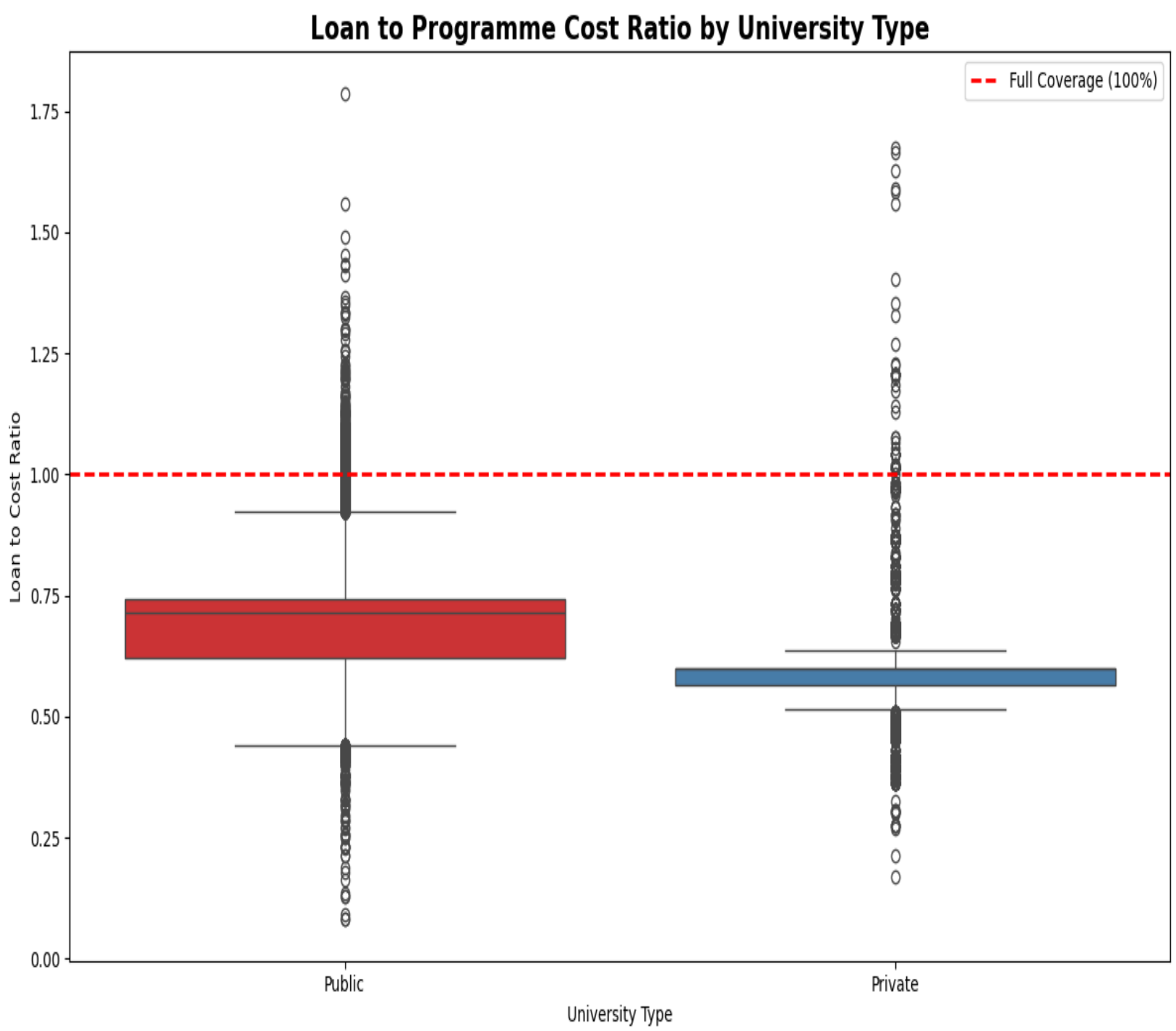
✓ National Average:

The national average dropout rate is approximately 30.2%.



Age Group vs Dropout

- Younger students (18-21) show higher dropout rates
- Non-traditional students (22+) show distinct patterns
- Age distribution peaks at 19-21 years

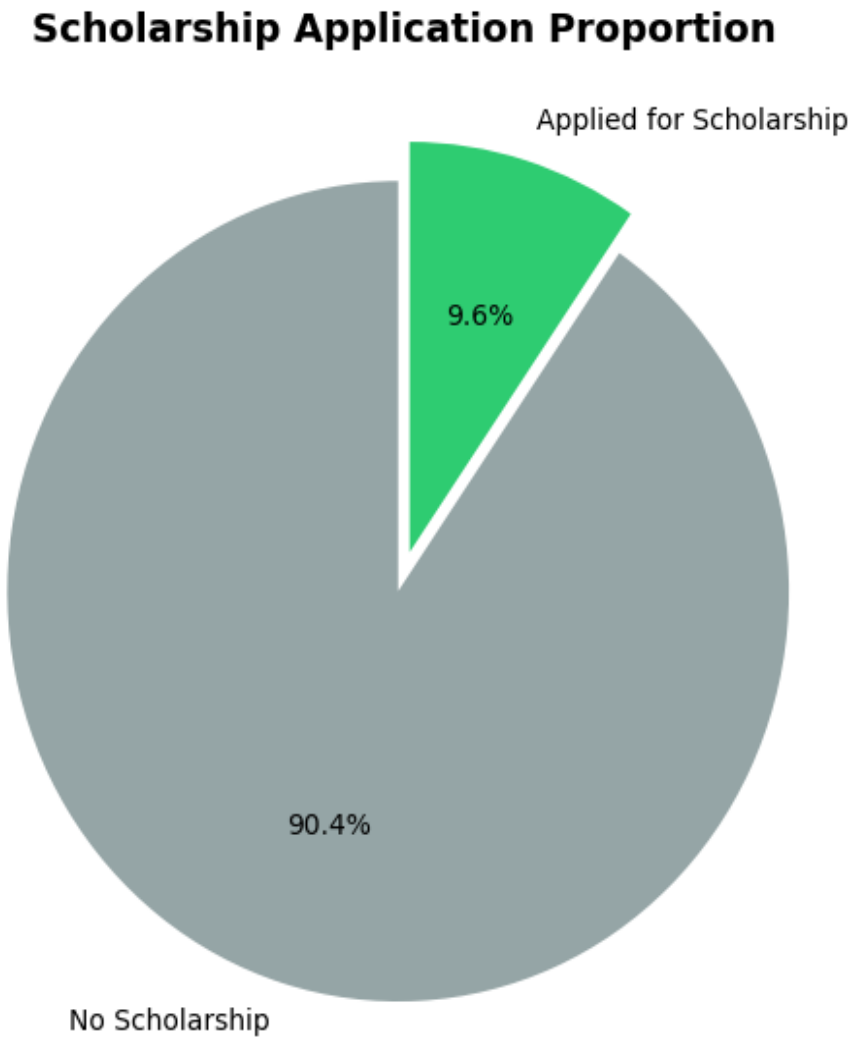


Dropout by Sponsorship & University Type:

Key insight

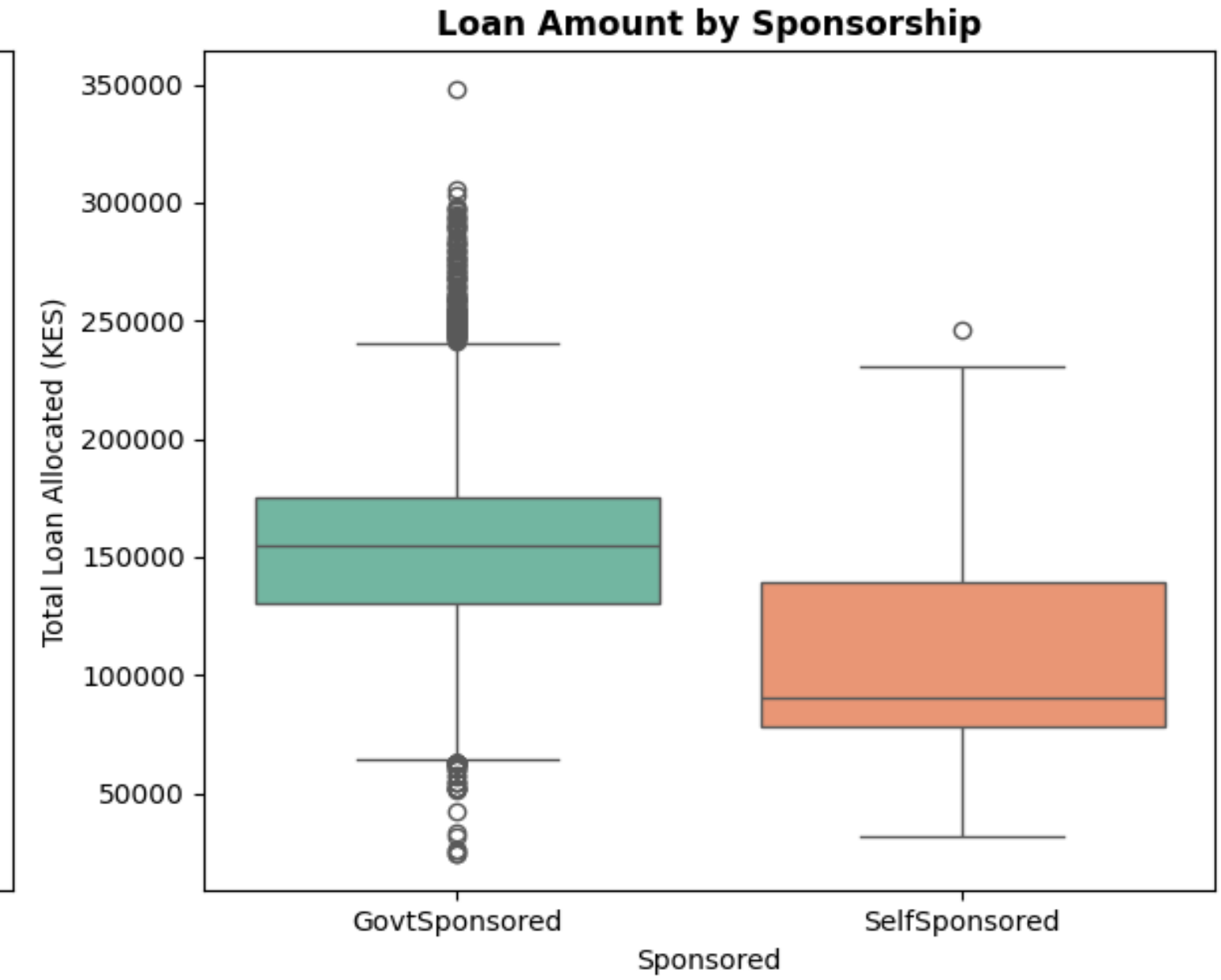
•Self-sponsored students face a much lower loan-to-cost ratio, indicating a higher financial burden.

•This financial strain correlates with a higher dropout rate at private institutions compared to public ones.



Loan Amount: Scholarship vs Non-Scholarship Applicants

Scholarship applicants receive higher median loan amounts, but no significant statistical association was found between scholarship application and dropout status ($p > 0.05$).



Loan Amount by Sponsorship

Self-sponsored students face a much lower loan-to-cost ratio, indicating a higher financial burden.

Predictive Modelling Approach

Target:

- ✓ Dropout (imbalanced, 69.85% Stayed vs. 30.15% Dropped).

Feature Engineering:

- ✓ new features were created to improve model performance by capturing hidden relationships in the data.

Preprocessing:

- ✓ Label Encoding for categorical features.
- ✓ Standard Scaling for Logistic Regression.

Class Imbalance Handling

- ✓ Used class weight='balanced' t during model training to give more importance to the minority (dropout) class.

Models Tested

1. **Logistic Regression** (Baseline)
2. **Random Forest (Tuned)**
3. **XGBoost (Tuned)**
4. **Balanced Random Forest** (Specifically designed for imbalanced data)
5. **Gradient Boosting**
6. **Stacking Ensemble** (Combination of base models)

Evaluation Metrics

- Accuracy: Overall correctness
- Precision: How many predicted dropouts were correct
- Recall: Ability to detect actual dropouts (very important here)
- F1-score: Balance between precision and recall
- ROC-AUC: Measures the area under the ROC curve. A higher AUC value indicates better model performance as it suggests a greater ability to distinguish between classes.

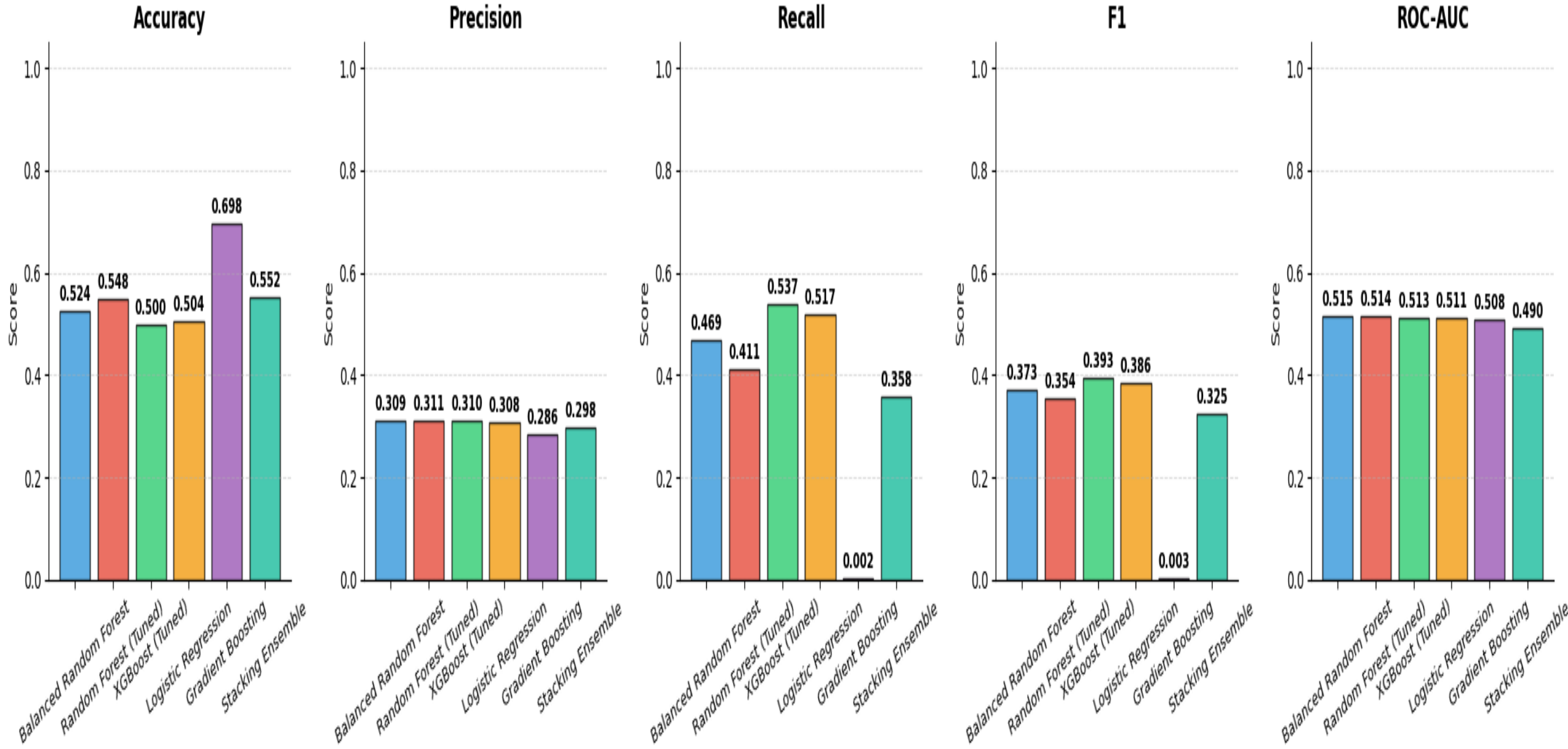
Note:

Recall is especially important because missing a dropout case means losing an opportunity for intervention.

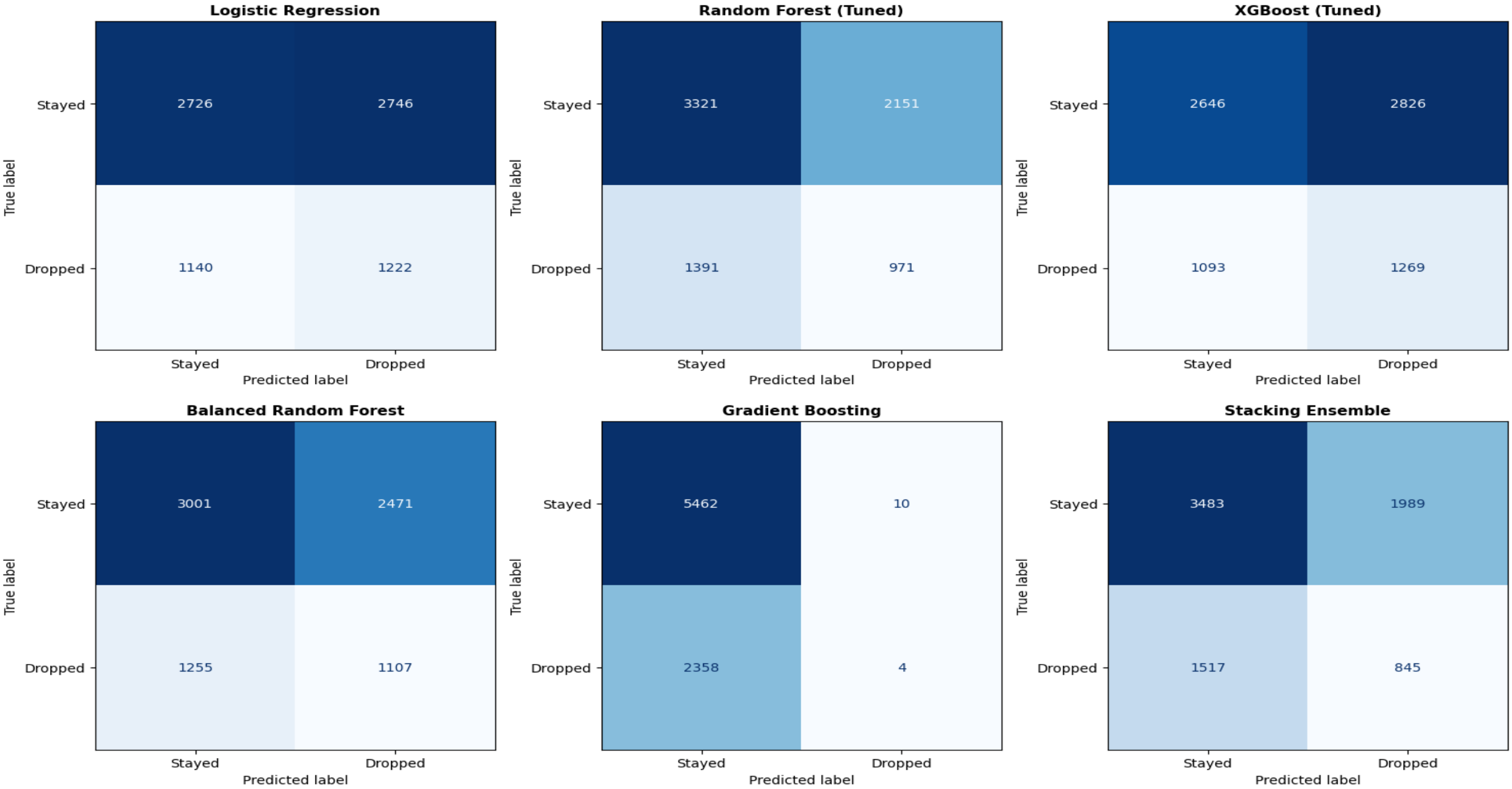
MODEL COMPARISON SUMMARY

Model	Accuracy	Precision	Recall	F1	ROC-AUC
XGBoost (Tuned)	0.4997	0.3099	0.5373	0.3931	0.5128
Logistic Regression	0.5040	0.3080	0.5174	0.3861	0.5109
Balances Random Forest	0.5244	0.3094	0.4687	0.3727	0.5150
Random Forest(Tuned)	0.5479	0.3110	0.4111	0.3541	0.5141
Stacking Ensemble	0.5525	0.2982	0.3577	0.3253	0.4902
Gradient Boosting	0.6977	0.2857	0.0017	0.0034	0.5077

Model Performance Comparison



Confusion Matrices



Best Model

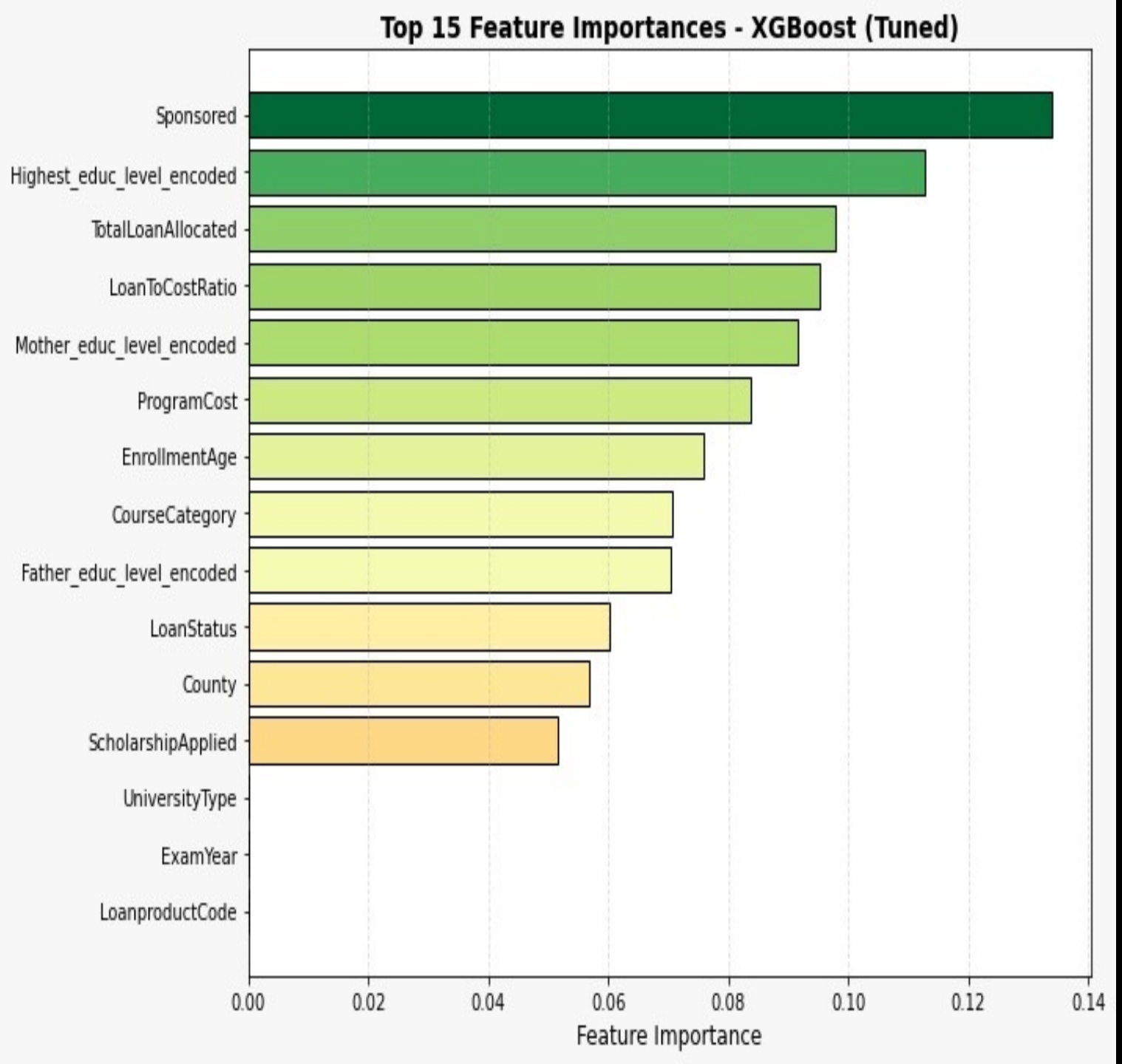
XGBoost(Tuned)

The **XGBoost(Tuned)** model achieved the highest recall (0.5373) making it the most effective at identifying students who eventually dropped out.

Conclusion:

The **XGBoost** is the best-performing model for this prediction task.

Feature Importance



Top Drivers:

1. Loan to cost ratio: The most important feature. A higher ratio (less loan covering cost) is a strong risk factor.

2. Institutional Factors: University type and Sponsored are key predictors.

3. Academic & Demographic: Course category, Enrollment age, and County also play significant roles.

4. Parental Education: Highest education level has moderate predictive power.

Interpretation:

Financial strain is the single most powerful predictor of dropout

Actionable Recommendations: For Policymakers & University Administrators:

1. Enhance Financial Aid Programs:

- ✓ **Action:** Target students with a high Loan to cost ratio (those whose loans cover a smaller portion of their fees).
- ✓ **Why:** This is the strongest predictor of dropout. Provide supplementary bursaries, work-study opportunities, or emergency funds.

2. Implement Early Warning Systems:

- ✓ **Action:** Use the model in real-time to flag students with high-risk profiles (e.g., high loan to cost ratio, specific course category, older enrollment age).
- ✓ **Why:** Allows for proactive academic advising, financial counseling, and targeted support before the student disengages.

3. Regional & Institutional Interventions:

- ✓ **Action:** Develop specific retention programs for high-risk counties (e.g., Mandera, Wajir) and for students in self-sponsored or private university programs.
- ✓ **Why:** Dropout risk varies significantly by geography and institutional type. Customized strategies will be more effective.

Recommended Next Steps:

✓ Feature Enhancement:

Incorporate new data sources like real-time academic performance (grades, attendance) and student engagement metrics (library usage, online portal logins) to improve the model.

✓ Pilot Intervention Programs:

Launch a pilot program based on the model's top risk factors (e.g., a targeted financial aid or academic coaching program) and measure its impact.

✓ Continuous Monitoring:

•Regularly retrain the model with new data to ensure its accuracy and relevance over time.

Deployment

The model is deployed using Streamlit.

- ✓ Streamlit provided a simple user interface for our predictions.

Live Application/Link;

- ➢ https://studentdropoutpredictionproject-ryuvm7jungzphlmptrntja.streamlit.app/

Thank you!